Task 1: Exploring and Visualizing the Iris Dataset

Objective

The objective of this task is to explore and visualize the Iris dataset using Python libraries such as Pandas, Matplotlib, and Seaborn.

Dataset Understanding

The Iris dataset contains measurements of iris flowers. The dataset includes the following features:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width
- Species

The dataset was loaded successfully using the Seaborn library.

Data Exploration

The dataset structure was examined using:

- Shape
- Columns
- First five rows of data

Dataset Understanding

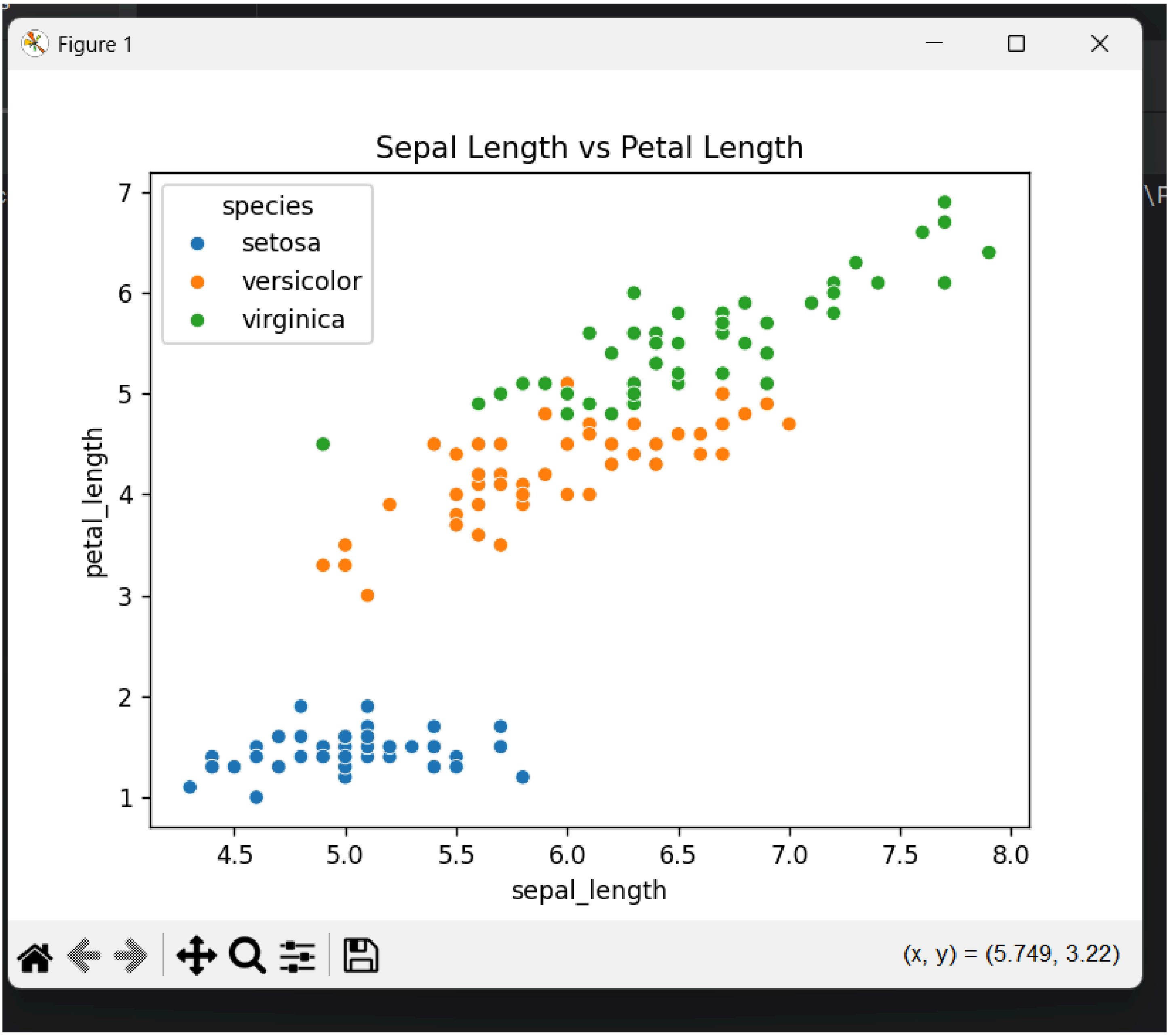
```
Shape: (150, 5)

Columns:
Index(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
       'species'],
      dtype='str')

First 5 Rows:
   sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa

Process finished with exit code 0
```

Figure 1: Dataset Structure and Preview

Observation:

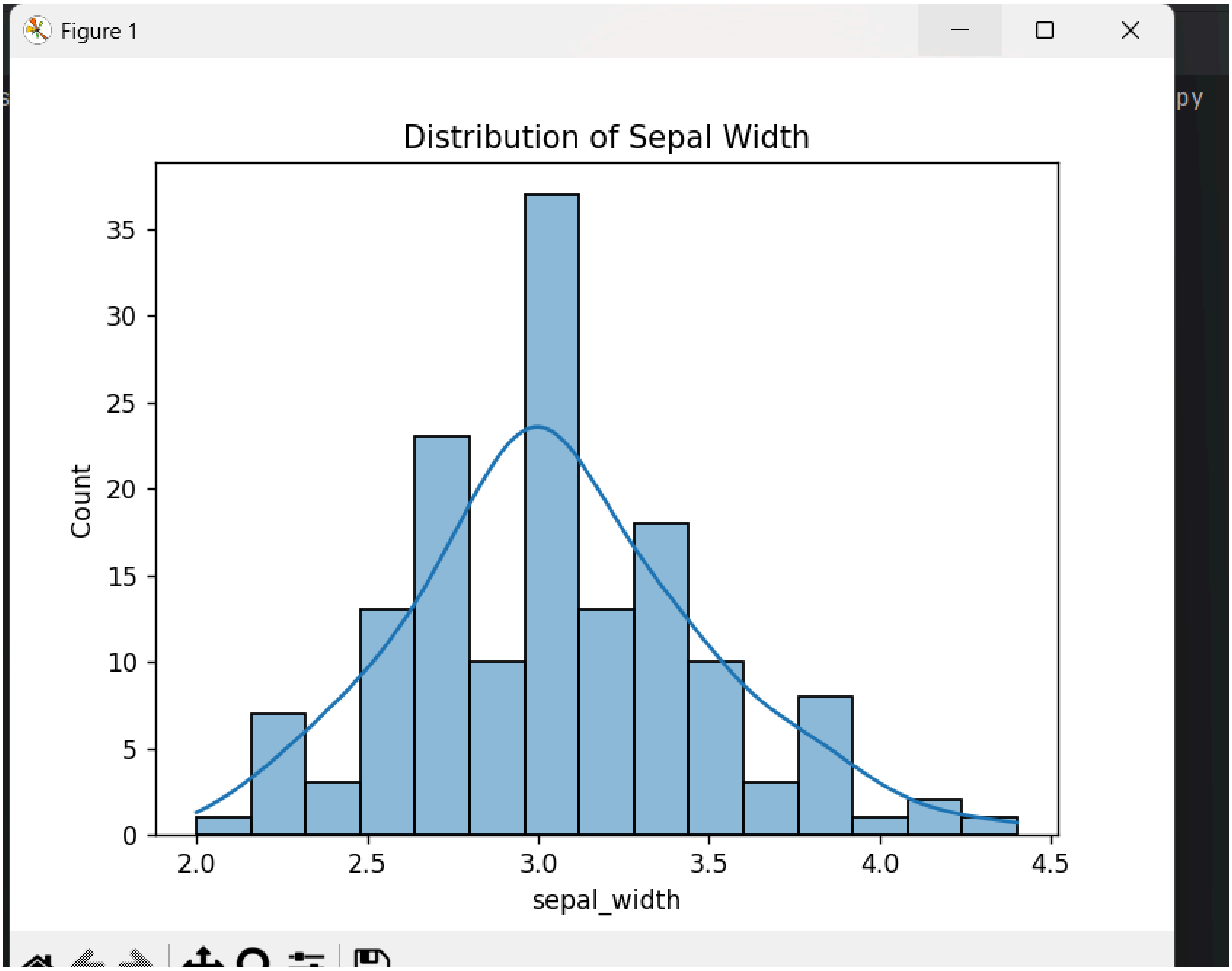
The Iris dataset contains 150 records and 5 columns. The features include sepal length, sepal width, petal length, petal width, and species. The first five rows provide a preview of the dataset structure and values.

Scatter Plot



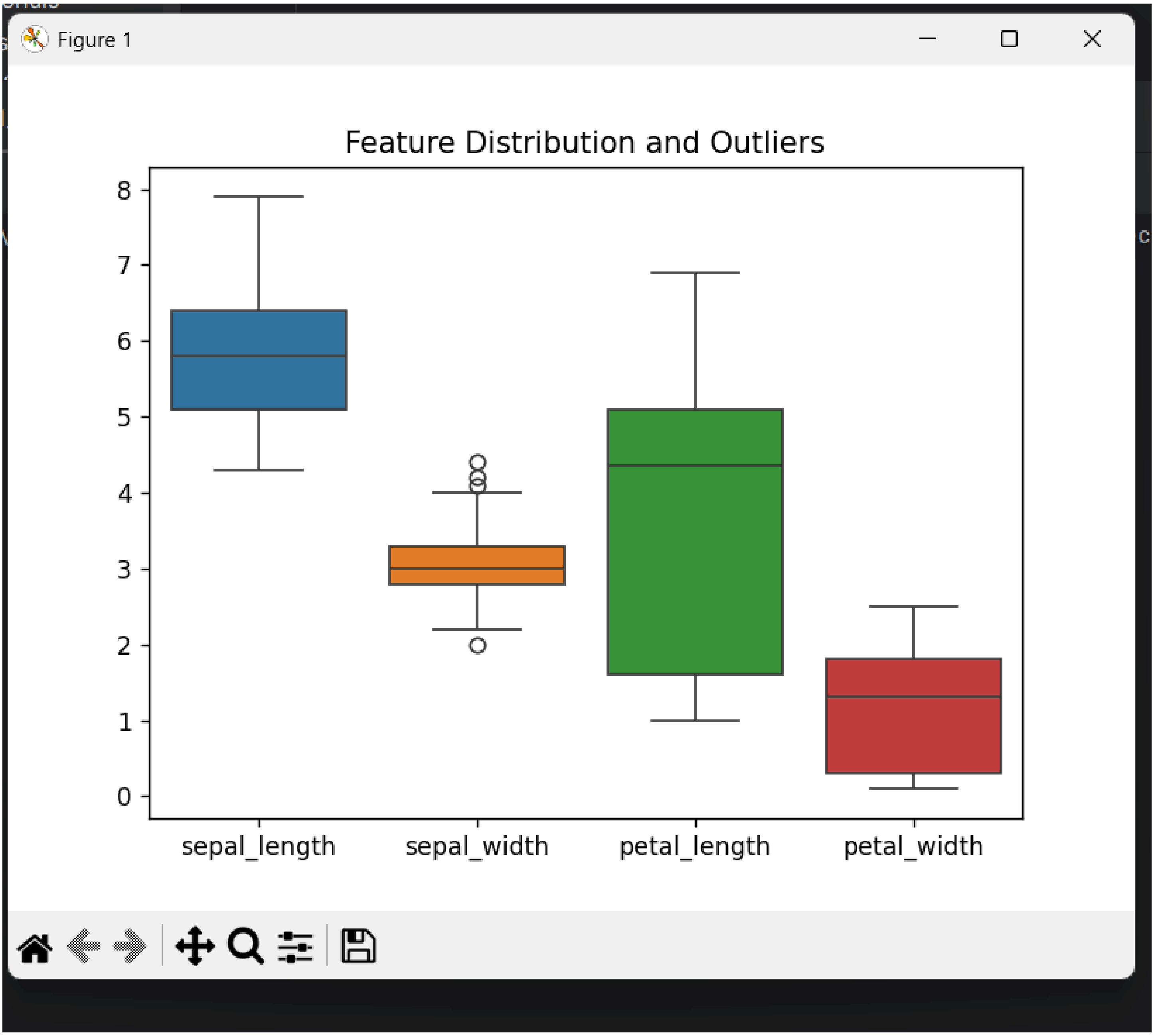
Observation: The scatter plot shows a relationship between sepal length and petal length. Different species form distinct clusters.

Histogram



Observation: The histogram shows that most sepal width values are concentrated around 3.0. The distribution is approximately bell-shaped, indicating that the majority of flowers have sepal widths close to the average value, with fewer flowers having very small or very large widths.

Box Plot



Observation: The box plot displays the distribution and spread of each feature in the Iris dataset. Sepal width contains a few outliers, while petal length shows the largest variation among the features. The plot helps identify differences in data spread and potential extreme values.

Conclusion

Conclusion

The Iris dataset was successfully explored and visualized using Python, Pandas, Seaborn, and Matplotlib. The dataset structure was examined using shape, columns, and sample records. A scatter plot was used to analyze relationships between variables, a histogram was used to study data distribution, and a box plot was used to identify variation and outliers. These visualizations provided valuable insights into the characteristics of the Iris dataset and demonstrated fundamental data analysis techniques.